

Power equations



1 Solve for the pronumeral for each of the following equations:

a $30^n = \sqrt[3]{30}$

b $3^y = \frac{1}{27}$

c $4^y = 64$

d $7^x = 1$

e $8^x = \frac{1}{8^2}$

f $(\sqrt{2})^k = 0.5$

g $\frac{1}{36} = 6^n$

h $(\sqrt{6})^y = 36$

i $5^x = \sqrt[3]{5}$

j $8^x = \sqrt{8}$

k $3^x = \sqrt[8]{3}$

l $4^x = 4^8$

m $x^5 = 3^5$

n $9^x = 9^{-4}$

o $3^x = 3^{\frac{2}{9}}$

2 Solve for x for each of the following equations:

a $x^{-7} = \frac{1}{6^7}$

b $7(4^x) = \frac{7}{4^3}$

c $10^x = 0.01$

d $5^{10x+33} = 125$

e $5^{-3x-1} = 3125$

f $(2^2)^{x+7} = 2^3$

g $(2^4)^{2x-10} = 2^2$

h $27(2^x) = 6^x$

i $9^y = 27$

j $3^{5x-10} = 1$

k $\frac{1}{3^{x-3}} = \sqrt[3]{9}$

l $30 \times 2^{x-6} = 15$

m $a^{x+1} = a^3 \sqrt{a}$

n $25^{x+1} = 125^{3x-4}$

o $\left(\frac{1}{8}\right)^{x-3} = 16^{4x-3}$

p $3^{x^2-3x} = 81$

q $\left(\frac{8}{3}\right)^x = \left(\frac{8}{3}\right)^7$

r $9^{x+3} = 27^x$

Polynomial equations

3 Solve for x for each of the following cubic equations:

a $x^3 - 125 = 0$

b $3x^3 = 5x^2$

c $x^3 = -8$

d $8x^3 - 125 = 0$

e $(x+8)(x+4)(1+x) = 0$

f $(5x-4)(x+3)(x-2) = 0$

g $x^3 - 49x = 0$

h $512x^3 - 125 = 0$

i $x^3 - 3x^2 - 18x + 40 = 0$

j $x^3 - 4x^2 - 45x = 0$

k $x^3 + 9x^2 + 27x + 27 = 0$

l $-64x^3 + 48x^2 - 12x + 1 = 0$

m $x^3 - 5x^2 - 49x + 245 = 0$

n $x^3 + 13x^2 + 47x + 35 = 0$

o $150x^3 + 115x^2 - 118x - 56 = 0$

p $729x^3 + 8 = 0$

q $x^3 - 5x^2 - 4x + 20 = 0$

r $x(x-5)(x+7) = 8(x-5)(x+7)$



4 Use technology to solve the following equations:

a $147x^3 + 427x^2 + 160x - 84 = 0$

b $50x^3 + 155x^2 + 152x + 48 = 0$

c $64x^3 + 40x^2 + 54x + 18 = 0$

d $48x^3 - 212x^2 + 84x + 209 = -36$

e $-576x^3 + 272x^2 - 41x + 78 = 76$

f $-25x^3 + 50x^2 - 19x - 27 = -45$

g $2x^4 + 5x^3 - 14x^2 - 5x + 12 = 0$

h $6x^4 + 9x^3 - 186x^2 + 315 = 0$

i $6x^5 - 5x^4 - 43x^3 + 70x^2 - 24x = 0$

j $5x^5 + x^4 - 25x^3 - 5x^2 + 20x + 4 = 0$

5 Solve for the roots of the cubic $P(x) = x^3 - 2x^2 - 5x + 6$ that has a factor of $x - 3$.